Subhasri Vijay

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EDUCATION

JOHNS HOPKINS UNIVERSITY, Baltimore, Maryland
Master of Science in Engineering: Computer Science — **Graduated May 2026**
Bachelor of Science: Computer Science; Minors: Computational Medicine, Entrepreneurship and Management — **Graduated May 2025**
Relevant Coursework: Data Structures, Computer System Fundamentals, Algorithms, Deep Learning, Artificial Intelligence, Computational Genomics: Sequencing, Databases, Human Computer Interaction, Computer Vision, Machine Translation

SPECIALIZED SKILLS

- **Machine Learning & AI**: PyTorch, TensorFlow, Deep Learning, Computer Vision, Foundation Models, Model Training & Evaluation
- **Programming Languages**: Python, Java, C/C++, JavaScript, TypeScript, SQL
- **Web Development**: React, TypeScript, Full-Stack Development, HTML/CSS
- **Data & Analysis**: NumPy, Pandas, Data Preprocessing, Dataset Curation, Jupyter
- **Tools & Platforms**: Git, Linux, MATLAB, R

PROJECTS AND EXPERIENCE

Cataract Research Assistant, Wilmer Eye Institute — January 2022-May 2026
- Created and analyzed six surgeon-specific ground-truth strategies to benchmark model performance on the question: "Can AI accurately identify the expert surgeon?"; evaluated using accuracy, sensitivity, specificity, and precision. Findings presented at AUPO 2024.
- Curated and preprocessed 30+ cataract surgery videos in collaboration with clinicians to build high-quality, structured datasets for training and evaluating deep learning models.
- Applied and evaluated deep learning architectures (encoders, decoders) and metrics (AUC, confusion matrix) for surgical skill assessment, contributing to clinically relevant advancements in AI-driven surgical evaluation.

Co-Founder, RescueReady — January 2025-Sept 2025
- Co-founded and led development of an AI-powered EMT training platform that simulates high-pressure emergency scenarios to enhance decision-making and situational awareness.
- Built a full-stack web application using React, TypeScript, and SQLite, integrating generative AI to generate realistic, protocol-aligned simulations in collaboration with EMTs and healthcare professionals.
- Designed RESTful API architecture and implemented user authentication system to support multi-user training sessions with progress tracking.
- Achieved 3rd Place at HopStart General Ventures and won the Pava Center for Entrepreneurship Award, responsible for product vision, technical architecture, and user testing.

IMAZER (Image Processing Tool) — July 2024
- Developed a lightweight Python desktop application for batch image resizing with a Tkinter GUI, eliminating command-line dependency for non-technical users.
- Engineered support for multiple image formats with robust error handling for invalid files; optimized batch processing pipeline for reliable performance across large image sets.

CAREATHON (Accessibility Tool) — July 2024
- Developed an accessibility tool using Python with speech-to-text, text-to-speech, and live translation capabilities to assist users with speech and hearing impairments.
- Leveraged advanced Python libraries for speech and language processing, contributing to inclusive healthcare communication and enabling expressive, multi-language interaction.

POSTERS AND PUBLICATIONS

- Poster titled "**Can AI accurately answer:" Who is the expert surgeon?"** at AUPO 2024.
- Poster titled "**Surgical Skill Assessment of Cataract Videos**" at ISBI 2025.
- Paper titled "**A Vision Foundation Model for Cataract Surgery Using Joint-Embedding Predictive Architecture**" at MIDL 2025.

TEACHING EXPERIENCE

Head Teaching Assistant, Data Structures: EN.601.226. — February 2024-May 2026
- Guided students in debugging complex programming tasks and simplified Data Structures concepts into accessible explanations, fostering better understanding, problem-solving, and technical communication skills.
- Conducted weekly office hours, providing hands-on assistance and promoting student engagement through practical learning experiences.
- Graded over 100 project submissions and 200 exam submissions while conducting 50+ office hours, ensuring fair grading and collaborating with faculty to create an inclusive and effective learning environment.

HONORS AND AWARDS

- Awarded Joel Dean Award for Excellence in Teaching by Computer Science department — 2026
- Dean's List for 8 semesters at Johns Hopkins University and graduated with General and Departmental Honors — 2025
- Awarded the Masson Fellowship for 2024 by the Computer Science department for Research. — 2025